

Backend and Model Comparison ? 2026-09-02

Executive result

This comparison now uses the benchmark that WebConsole actually uses: the CLI transport. Raw HTTP is not an active WebConsole configuration and is included only in the diagnostic appendix because it exposed a separate gateway problem.

The benchmark ran 14 tasks, 2 repeats per task, with mechanical execution for code and claim checks for prose. Results below come from 264 measured CLI runs:

? 4 Anthropic models ×14 tasks ×2 repeats = 112 runs

? 5 Azure models ×14 tasks ×2 repeats = 140 runs

? Qwen ×6 floor/simple tasks ×2 repeats = 12 runs

Qwen was intentionally re-tested on simpler tasks. The clean or empty output seen in the old benchmark was a harness failure, not evidence that Qwen was unusable. Qwen is self-hosted and free; Anthropic models have the measured token costs below; Azure prices were not recorded.

Recommendation by scarce resource

Constraint | First choice | Why
Lowest cost | vllm/Qwen3.6-35B-A3B-NVFP4 | 11/12 on the floor/simple CLI tier, 5.2 s median, \$0 marginal token cost
Best measured quality | claude-fable-5 | 28/28 CLI runs correct; most expensive Anthropic option
Best Anthropic value | claude-haiku-4-5 | 27/28, \$0.0433 per correct answer, fastest Anthropic TTFT
Fast capable Azure option | azureai/gpt-5.4-mini-copilot | 24/28, 5.8 s median; Azure price not recorded
Fastest measured Azure option | azureai/gpt-5.4-mini | 25/28, 6.3 s median; Azure price not recorded
Higher-quality Azure fallback | azureai/gpt-5.6-sol | 25/28, but 10.5 s median and Azure price not recorded

No single quality-only ranking should hide cost. Qwen is the correct default when latency is not scarce. Anthropic is the expensive quality tier. Azure cannot be ranked on cost until its deployment rate card is recorded.

CLI measurements

All rows below are CLI runs. Correct means all core checks for the run passed. Edge checks remain visible in task scores but do not decide whether the task is solved. \$ total is the CLI-reported cost only when the result carries costBasis: list; unknown Azure pricing is reported as n/r, never as zero.

Tier	Model	Runs	Correct	Median total	Median TTFT	Median output tokens	\$ total	\$/correct
Anthropic	claude-fable-5	28	28/28	9.1 s	3.31 s	142	\$7.3130	\$0.2612
Anthropic	claude-opus-5	28	27/28	8.3 s	2.55 s	122	\$4.8428	\$0.1794
Anthropic	claude-sonnet-5	28	27/28	6.0 s	1.97 s	132	\$2.4518	\$0.0908
Anthropic	claude-haiku-4-5	28	27/28	10.6 s	1.75 s	611	\$1.1682	\$0.0433
Azure	azureai/gpt-5.6-sol	28	25/28	10.5 s	4.80 s	131	n/r	n/r

What the table means

? vllm/Qwen3-0.6B was requested but could not be benchmarked: it does not appear in aimachines.activemodels for the CF AI Machine backend (bin/wc-backend-env.py --resolve-model returns nothing), so the CLI would route it as a 429 wearing a capacity error's clothes rather than a real answer. Needs a corrected model id or registration before it can be measured.

CLI task comparison

Task Difficulty Verified by Fable Opus Sonnet Haiku gpt-5.6-sol
gpt-5.4-mini gpt-5-mini copilot luna Qwen
floor-add floor execution 100 100 100 100 100 100 100 100
100 100
simple-fizzbuzz simple execution 100 100 100 100 100 100 100
100 100 100
simple-count-vowels simple execution 100 100 100 100 100 100 100
100 100 100 100
simple-reverse-words simple execution 100 100 100 100 100 100 100
100 100 100 100
simple-sum-evens simple execution 100 100 100 100 100 100 100
100 100 100 100
simple-json-field simple execution 75 75 75 75 75 100 100
75 75 75/0
coding-bug-fix hard execution 100 100 94 100 100 100 100
100 100 ?
coding-algo hard execution 100 100 96 98 100 100 100 100
100 ?
reasoning-puzzle hard claim 100 100 75 100 50 100/50 50 50
50 ?
reasoning-math hard claim 100 100 100 75/100 100 100 100

100 | 100 | ?
comprehension-read | hard | claim | 100 | 100/75 | 100 | 100 | 75/100 | 75 |
75/100 | 75 | 75 | ?
planning-task | hard | claim | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
100 | ?
long-context-needle | hard | claim | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
100 | 100 | ?
multi-turn-resume | hard | execution | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
100 | 100 | ?

Qwen's two CLI repeats on simple-json-field scored 75 and 0: one run passed the core checks but failed the optional edge check, while the other omitted the json import and failed execution. This is the one measured Qwen miss, not a clean/empty response.

Task-level observations

? All models handled the five simplest executable tasks consistently.

? simple-json-field exposes a common idiom failure: direct indexing of an optional field instead of using .get(). This is an edge-quality signal, not a reason to reject a model that passes the core task.

? The malformed reasoning-puzzle target asks for 4 litres in a 3-litre jug. Models that identify the impossibility receive only the claim-check partial score. This task is useful for constraint awareness but should not be read as evidence that the models failed to reason.

? The hard coding tasks separate the Anthropic tier slightly from the Azure tier, but the sample is small and consists of only two repeats.

Consistency

"Unstable" means the same prompt produced different scores in its two repeats.

Model / transport | Unstable tasks
claude-fable-5 / CLI | 0 of 14
claude-opus-5 / CLI | 1 of 14
claude-sonnet-5 / CLI | 3 of 14
claude-haiku-4-5 / CLI | 2 of 14
azureai/gpt-5.6-sol / CLI | 1 of 14
azureai/gpt-5.4-mini / CLI | 1 of 14
azureai/gpt-5-mini / CLI | 1 of 14
azureai/gpt-5.4-mini-copilot / CLI | 1 of 14
azureai/gpt-5.6-luna / CLI | 0 of 14
vllm/Qwen3.6-35B-A3B-NVFP4 / CLI | 1 of 6

Two repeats show that instability exists; they do not estimate long-run failure rates. Sonnet's three unstable tasks include a 50-versus-100 swing on reasoning-puzzle, so one-shot results should not be treated as deterministic.

Cost

Recorded rate card

Model/backend | Input / 1M tokens | Output / 1M tokens | Cache read | Cache write | Basis
claude-fable-5 | \$10.00 | \$50.00 | \$1.00 | \$20.00 | published
claude-opus-5 | \$5.00 | \$25.00 | \$0.50 | \$10.00 | published and independently confirmed
claude-sonnet-5 | \$2.00 | \$10.00 | \$0.20 | \$4.00 | published
claude-haiku-4-5 | \$1.00 | \$5.00 | \$0.10 | \$2.00 | published
vllm/Qwen3.6-35B-A3B-NVFP4 | \$0 | \$0 | ? | ? | self-hosted, marginal token cost treated as zero

azureai/ | not recorded | not recorded | not recorded | not recorded | no
deployment rate supplied

Anthropic totals use the CLI's totalcostUSD only when costBasis is list. The harness also captures uncached input, cache-read, cache-write, and output tokens. Unknown vendor pricing is not silently converted to zero.

Why old non-Anthropic console costs are not usable

The old usage table charged non-Anthropic backends with an Anthropic fallback rate. That made self-hosted Qwen appear to cost money and produced fictional Azure costs. Those figures are excluded from this comparison. Qwen is free under the stated self-hosted deployment assumption. Azure remains n/r until its actual rate card is supplied.

Qwen: corrected interpretation

The first benchmark ranked Qwen at 55.4% and described empty output as a model failure. That result is withdrawn.

The old harness used a 4,096-token cap for both reasoning and answer, a 180-second timeout, and an extractor that collapsed truncated thinking into the literal [thinking] marker. Qwen's reasoning output consumed the budget and time limit. A truncated answer and an empty answer became indistinguishable.

The corrected benchmark used 16,384 maximum output tokens, a 1,000-second timeout, private per-run directories, file-delivery recovery, and explicit core versus edge checks. The simpler CLI tier produced:

? 11/12 core tasks correct
? 5.2 s median total time
? 0.74 s median TTFT
? 56 median output tokens
? \$0 marginal token cost
? no clean/empty runs in the corrected CLI result

The one miss was simple-json-field, where one answer omitted import json. That is a real answer defect. It is not the earlier harness artefact.

Historical HTTP diagnostic: not active configuration

Raw HTTP was removed from the benchmark because WebConsole production uses the CLI path. The following measurements remain only to explain why old conclusions were wrong.

Qwen head-to-head, same six floor/simple tasks

Metric | HTTP | CLI
Correct | 11/12 | 11/12
Median total time | 88.2 s | 5.2 s
Median output tokens | 2,134 | 56
Median thinking characters | 6,540 | 102
Median TTFT | 76.71 s | 0.74 s

The model's correctness was the same. The HTTP path generated far more reasoning and was much slower. Those HTTP figures must not be used to choose the current WebConsole backend because that transport is no longer part of the application benchmark.

Azure HTTP gateway failure

The historical Azure HTTP stream returned a LiteLLM gateway error (list index out of range) for most streaming attempts. The stream error carried no normal type field, so the old parser recorded silence. The corrected parser surfaces the error and can re-run unstreamed, but the raw HTTP transport was then removed from the production comparison.

This is a gateway/transport finding, not a model-quality finding. It should be fixed in the gateway separately if HTTP streaming is needed. It does not change the CLI measurements above.

Benchmark method and limits

- ? Harness: bin/wc-bench.py
- ? Report generator: bin/wc-bench-report.py
- ? Verifiers: bench/verify.py
- ? Tasks: bench/tasks.py
- ? Cost logic: bench/cost.py
- ? Maximum output tokens: 16,384
- ? Timeout: 1,000 seconds
- ? Repeats: 2 for the completed comparison
- ? Active transport: CLI only
- ? Code correctness: extracted code executed in a subprocess with a restricted environment
- ? Prose correctness: mechanical claim checks, weaker than execution
- ? Core checks decide correct; edge checks remain in the score and report
- ? Result files: benchanthropic20260902.json, benchazure20260902.json, benchqwentransport20260902.json

Limitations remain:

- ? Azure and Anthropic are not identical vendor deployments, so raw scores do not establish general model rankings outside this environment.
- ? Qwen was re-run on the floor/simple tier after the original harness failure; its 11/12 is not a 14-task score.
- ? Two repeats are enough to expose some instability, not enough to establish production failure probabilities.
- ? Claim checks can miss errors that execution would catch.
- ? Azure rates are not recorded, so no Azure cost-per-correct result is claimed.
- ? The benchmark measures model responses in isolated CLI runs, not complete end-to-end WebConsole user journeys.

Files and provenance

File | Contents
benchanthropic20260902.json | 112 Anthropic CLI runs
benchazure20260902.json | Completed Azure run: 280 historical CLI/HTTP rows, including diagnostic HTTP rows
benchqwentransport20260902.json | 12 Qwen CLI rows and 12 historical HTTP rows
benchqwen350.8bfloor20260903.json | 12 vllm/Qwen3.5-0.8B CLI rows, floor/simple tier
benchqwen350.8b20260903.json | Partial (20-row) hard-tier attempt, killed after reasoning-puzzle hit 1130.96s ? kept for the record, not scored into the table above
benchrates.json | Explicit rate card used by the harness
bin/wc-bench-report.py | Reproducible report generator

The final recommendation is therefore conditional, not a single league-table number:

? Choose Qwen for free background and batch work where the corrected floor/simple task bar is sufficient.

? Choose Haiku for the best measured Anthropic cost/quality balance.

? Choose Fable when the extra clean-sweep quality is worth its much higher cost.

? Choose Azure based on latency and task fit only until actual Azure pricing is recorded.